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FACULTY OF ENGINEERING AND GREEN TECHNOLOGY

UGEA3233 – ANALOGUE CMOS IC DESIGN

ASSIGNMENT

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Results

Discussion

AC Analysis

The AC Analysis is based on this following schematic diagram:

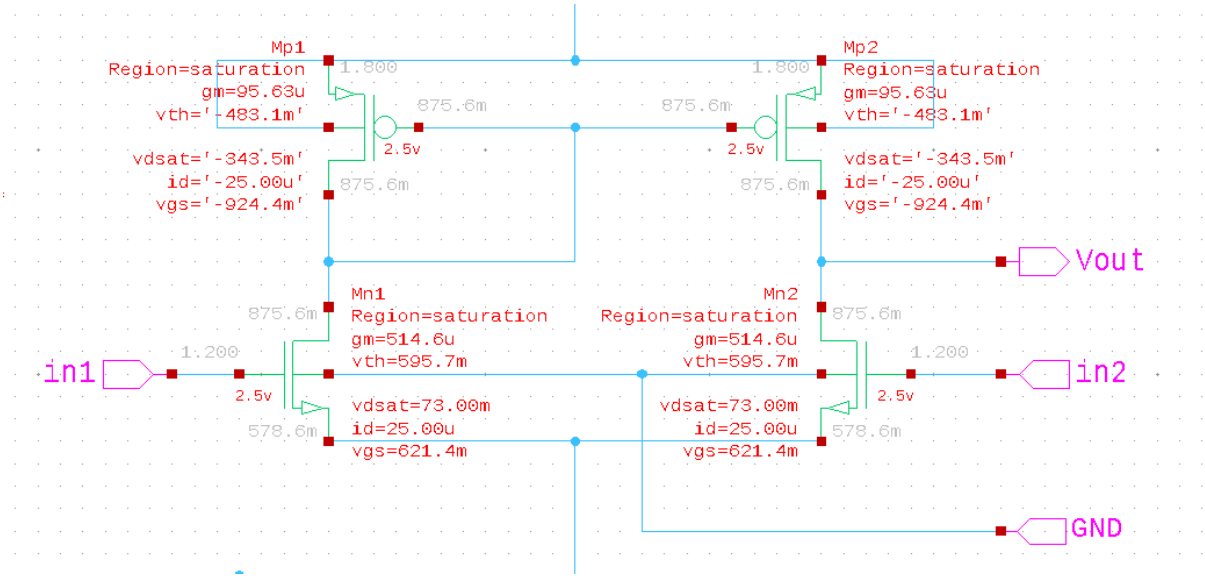


Figure ___ : Schematic Diagram of Differential Amplifier

Several assumptions is made, the dimensions of the PMOS Mp1 (M3) and PMOS Mp2 (M4) is the same when NMOS Mn1(M1) and NMOS Mn2(M2) shares the same dimensions

In order for better signal analysis, the schematic can be redrawn as a small signal circuit shown below:

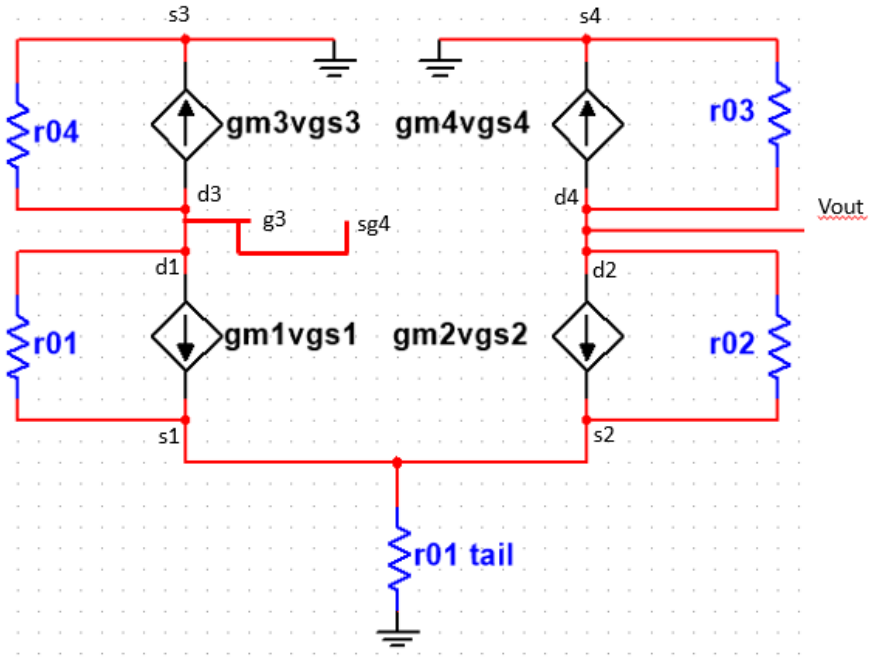


Figure ___ : Small Signal Circuit Diagram of Differential Amplifier

The analysis of the current source is omitted for the AC analysis and is treated as an independent premade current source. The specific details of current source analysis will be done later at the second section of the discussion.

AC Analysis: Differential Mode

Several key variable can be obtained from the measurement of the schematic diagram.

$$g_{m1} = g_{m2} = 514.6u, g_{m3} = g_{m4} = 95.63u$$

$$i_{d1} = i_{d2} = i_{d3} = i_{d4} = 25uA$$

Hence the following relationship can be obtained:

$$r_{01} = r_{02} \text{ and } r_{03} = r_{04}$$

By instructed, Kirchoff Current Law is used in the Differential Mode Analysis

At V_{out}:

$$gm_2V_{gs2} + gm_1V_{gs1} + \frac{V_{out} - V_{s2}}{r_{02}} + \frac{V_{out}}{r_{04}} = 0$$

Provided that: $V_{gs2} = (V_{in(-)} - V_{s2})$,

Equation 1: $gm_2(V_{in(-)} - V_{s2}) + gm_1V_{gs1} + \frac{V_{out}}{r_{02}// r_{04}} - \frac{V_{s2}}{r_{02}} = 0$

At d1

$$gm_1V_{gs1} + gm_3V_{gs3} + \frac{V_{d1}}{r_{03}} + \frac{V_{d1} - V_{s1}}{r_{01}} = 0$$

Provided that: $V_{gs1} = (V_{in(-)} - V_{s1})$,

Equation 2: $gm_1(V_{in(-)} - V_{s1}) + gm_3V_{gs3} + \frac{V_{out}}{r_{01}// r_{03}} - \frac{V_{s1}}{r_{01}} = 0$

Since $r_{01} = r_{02}$ and $r_{03} = r_{04}$ and $g_{m1} = g_{m2}, g_{m3} = g_{m4}$, hence the following can be obtained:

From 1: $gm_1(V_{in(-)} - V_{s1}) + gm_1V_{d1} + \frac{V_{out}}{r_{01}// r_{03}} - \frac{V_{s1}}{r_{01}} = 0$ → **Equation 3**

From 2: $gm_3(V_{in(-)} - V_{s1}) - gm_3V_{d1} + \frac{V_{d1}}{r_{01}// r_{03}} - \frac{V_{s1}}{r_{01}} = 0$ → **Equation 4**

By subtracting 3 by 4

$$g_{m1}V_{in} = \frac{V_{out}}{r_{01}// r_{03}}$$

$$A_v = \frac{V_{out}}{V_{in}} = g_{m1}\, r_{01}// r_{03}$$

Since

$$r_{01} = \frac{1}{\lambda pI_D} \text{ and } r_{03} = \frac{1}{\lambda nI_D}$$

Given that I_d = 0.025mA, and assuming λn be 0.225 when λp to be approximately 0.86 the following calculation can be done

$$r_{01} = r_{02} = \frac{1}{\lambda p I_D} = 177\text{kOhm}\ r_{03} = r_{04} = \frac{1}{\lambda n I_D} = 46.5kOhm$$

Hence:

$$A_v = g_{m1}\, r_{01} // r_{03}$$

$$A_v = 514.6 \times 10^{-6}\left(\frac{1}{177k} + \frac{1}{46.6k}\right)^{-1}$$

$$A_{vcalculated} = 18.95$$

The following value can be obtained in Differential Mode : V_{out}=376.75mV, $V_{in(+)}$=10mV, $V_{in(-)}$

$$A_{vc} = \frac{V_{out}}{V_{in(+)} - V_{in(-)}} = \frac{551.4u}{10mV - (-10mV)}$$

$$A_{v(actual)} = 18.8375$$

As the required specifications is to have a Differential Mode amplification that is larger than 10, our design meet the specifications.

$$Percentage\ Error = \frac{A_{v(actual)} - A_{vcalculated}}{A_{vcalculated}} = 0.59\%$$

The slight difference between the actual value and the calculated value might be due to the internal resistance of wire, as the difference is small it can be neglected.

Common Mode Analysis

Reusing the schematic from the above, the following can be obtained.

At Node V_{d3}: $g_{m3}\, V_{gs3} + g_{m1} V_{gs1} + \frac{V_d 3}{r_{o3}} + \frac{V_d 1}{r_{o1}} = 0$

At Node V_{out}: $g_{m2}\, V_{gs2} + g_{m4} V_{gs4} + \frac{V_d 2}{r_{o2}} + \frac{V_d 4}{r_{o4}} = 0$

At Node S_1: $-g_{m1} V_{gs1} - g_{m2} V_{gs2} + \frac{V_{s1} - V_{d1}}{r_{o1}} + \frac{V_{s2} - V_{out}}{r_{o2}} = 0$

At Node S_2: $-g_{m1} V_{gs1} - g_{m2} V_{gs2} + \frac{V_{s1} - V_{d1}}{r_{o1}} + \frac{V_{s2} - V_{out}}{r_{o2}} = 0$

Since M_1=M_2 hence V_{s1}=V_{s2}, gm1=gm_2, r_{01}=r_{02} and V_{g1}=V_{g2}=$V_{input(closed\ loop)}$=V_{ic}, V_d=V_{out}

$$-g_{m1}(V_g - V_{s1}) - g_{m1}(V_g - V_{s1}) + \frac{V_{s1} - V_{out} + V_{s1} - V_{out}}{r_{o1}} = 0$$

$$-2g_{m1} V_g + 2g_{m1} V_{s1} = -\frac{2V_{s1} + 2V_{out}}{r_{o1}}$$

$$r_{o1}(-2g_{m1} V_g + 2g_{m1} V_{s1}) = -2V_{s1} + 2V_{out}$$

$$-2g_{m1}V_g r_{01} + 2g_{m1}V_{s1}r_{o1} = -2V_{s1} + 2V_{out}$$

Equation 5: $V_{s1} = \frac{V_{out}+g_{m1}V_{ic}r_{o1}}{g_{m1}r_{o1}+1}$ can be obtained.

Provided that M_3=M_4 hence V_{s3}=V_{s4}, gm_3=gm_4, r_{03}=r_{04}

From **Equation 2** : $gm_1(V_{in(-)} - V_{s1}) - gm_3V_{gs4} + \frac{V_{out}}{r_{01}// r_{03}} - \frac{V_{s1}}{r_{01}}$
Since closed loop gain, V_{s1}=0, V_{gs4}=V_{out} hence

Equation 6: $gm_1(V_{in(-)} - V_{s1}) - gm_4V_{out} + \frac{V_{out}}{r_{01}// r_{03}}$=0

Combining **Equation 5** and **Equation 6:**

$$-gm_4V_{out} + gm_1\left(V_{ic} - \frac{V_{out} + -gm_1V_{ic}r_{o1}}{-gm_1r_{o1} + 1}\right) = 0$$

$$V_{out}(-gm_4)(gm_1r_{o1} + 1) + gm_1V_{ic}(gm_1r_{o1} + 1) - gm_1(V_{out} + gm_1V_{ic}r_{o1}) = 0$$

$$A_{vc} = \frac{V_{out}}{V_{ic}} = \frac{gm_1}{gm_3gm_1r_{o1} + gm_3 + gm_1}$$

Recalling: $g_{m1} = g_{m2} = 514.6u, g_{m3} = g_{m4} = 95.63u$

$$A_{vc} = \frac{514.6u}{95.63u \times 514.6u \times 177k + 514.6u, +95.63u}$$

$$A_{closed\ loop(calculated)} = 0.0552$$

From the observed value from the graph

The following value can be obtained in Common Mode: V_{out}=551.4μV, V_{ic} = 10mV

$$A_{closed\ loop(actual)} = \frac{V_{out}}{V_{ic}} = \frac{551.4u}{10mV}$$

$$A_{closed\ loop(actual)} = 0.05514$$

$$Percentage\ Error = \frac{A_{closed\ loop(actual)} - A_{closed\ loop(calculated)}}{A_{closed\ loop(calculated)}}$$

$Percentage\ Error$=0.1%

The slight difference between the actual value and the calculated value might be due to the internal resistance of wire, as the difference is small it can be neglected.

Common Mode Rejection Ratio CMRR

$$CMRR = \frac{Differential\ Gain}{Common\ Mode\ Gain}$$

Utilizing the value in previous calculations:

$A_{v(actual)} = 18.8375$ and $A_{closed\ loop(actual)} = 0.05514$

$$CMRR_{Actual} = \frac{A_{v(actual)}}{A_{closed\ loop(actual)}}$$

$$CMRR_{Actual} = \frac{18.8375}{0.05514}$$

$$CMRR_{Actual} = 341.6304$$

Converting to values in dB

$$CMRR_{Actual} = 20\log(341.6304)$$

$$CMRR_{Actual} = 50.6711dB$$

As the required specifications is to have a CMRR that is larger than 50, our design meet the specifications.

$A_{closed\ loop(calculated)} = 0.0552$ and $A_{vcalculated} = 18.95$

$$CMRR_{calculated} = \frac{A_{v(calculated)}}{A_{closed\ loop(Calculated)}}$$

$$CMRR_{calculated} = \frac{18.95}{0.0552}$$

$$CMRR_{calculated} = 343.297$$

Converting to values in dB

$$CMRR_{calculated} = 20\log(343.297)$$

$$CMRR_{calculated} = 50.713dB$$

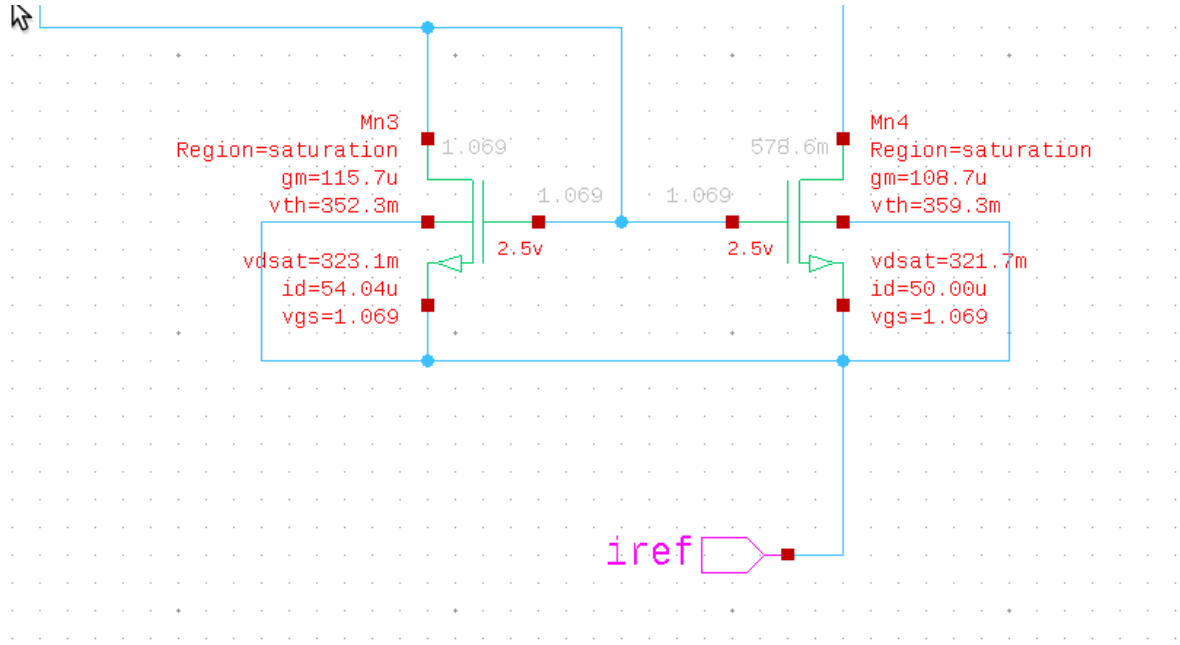
Percentage Error:

$$Percentage\ Error = \frac{CMRR_{Actual} - CMRR_{calculated}}{CMRR_{calculated}}$$

$Percentage\ Error$=0.083%

The percentage error is low to an extend that it can be ignored and be taken as the decimal calculation inconsistency. The nature of the percentage error to be lesser than 0.1% symbolizes that our design not only meet the specifications of the assignment requirements, it also has a low percentage error rate. Hence showing that it is a good design.

Current Analysis:



Assuming Mn3 be M5 and Mn4 be M6, M5 and M6 has the same dimensions of W_5=W_6=350n and L_5=L_6=250n , hence I_{d5}=I_{d6}=I_{ref}, this is proven by I_{d5} and I_d has similar value of 54u and 50u respectively, from observation of the schematic measurement, V_{GS5} =V_{GS6}=1.069V

Assuming that V_{thn} to be typical 0.8V, Typical value of KP_n ranges from 0.1 to 0.0001A , as we are using 250nm setting which is quite large, taking KP_n value to be 0.001A = 10uA

$$I_{d5} = \frac{KP_n}{2}\frac{W_5}{L_5}(V_{GS5} - V_{THN})^2$$

$$I_{d5} = \frac{0.001}{2}\frac{350n}{250n}(1.069 - 0.8)^2$$

$$I_{ref=}I_{d5} = 50.65 \times 10^{-6}$$

From direct reading of the value from the schematic, I_{d6} = I_{ref} = 50uA = 50x10^{-6}

$$Percentage\ Error = \frac{I_{ref(Actual)} - I_{ref(calculated)}}{I_{ref(calculated)}}$$

$$Percentage\ Error = 1.28\%$$

Unity Gain Frequency

Unity gain frequency can be obtained when the gain at one frequency is unity, hence the gain is 1.

Using Kirchoff Current Law,

$$I_{tail} = 0.05mA = 50uA$$

$$I_d = 0.5I_{ref} = 25uA$$

Assuming KPn to be 0.1, KP_{n1} differs from the KP_{n5} as M1 and M5 are of different settings. When W_1 is preset to be 1µ and L_1 to be 250n

The equation of gm can be obtained as below:

$$gm_1 = \sqrt{2 \times 0.1 \times \frac{W_1}{L_1} \times 25 \times 10^{-6}}$$

$$gm_1 = \sqrt{2 \times 0.1 \times \frac{1u}{250n} \times 25 \times 10^{-6}}$$

$$gm_1 = 0.0167332$$

The equation of Unity Gain frequency can be obtained as such

$$f = \frac{gm_1}{2\pi C_{load}}$$

When C_{load} is 1pF as required by the assignment guidelines

$$f = \frac{0.0167332}{2\pi 1pF}$$

$$f = 2663171494 = 2.663GHz$$

$$f_{calculated} = 2.663GHz$$

From observation, the unity gain frequency occur at 2.93297GHz

$$Percentage\ Error = \frac{f_{actual} - f_{calculated}}{f_{calculated}}$$

$$Percentage\ Error = 10.13\%$$

By far this is the highest percentage error that we observed, but it is still fix at around 10% which is the acceptable range of performance. We suspect the reason behind such phenomena is because the propagation delay that ultimately affect the frequency gain at high frequencies.